

# **Market Power**

**ECON 201: Principles of Microeconomics**  
**Lecture 12**

Div Bhagia

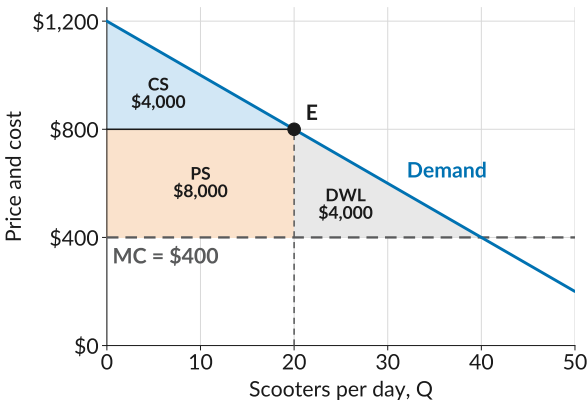
## Electric Scooters

- The company sells 20 scooters a day at \$800, at  $E$ . Each one costs \$400 to make.
- The three areas:

$$CS = \frac{1}{2} \times 20 \times 400 = 4,000$$

$$PS = 400 \times 20 = 8,000$$

$$DWL = \frac{1}{2} \times 20 \times 400 = 4,000$$



Why can this company charge more than marginal cost, and how much more?

## Market Power

- The scooter company sells a product that is at least a little different from what other firms sell.
- Some buyers prefer its scooters, so they do not all switch to another brand when it charges a little more.
- If every buyer switched to the cheapest scooter, a rival could charge a little less, and this company would sell nothing at \$800.
- We say a firm has **market power** if it can set its price above marginal cost without losing all its buyers to rivals.
- A firm whose product has few close **substitutes** has market power.

## Price Markup

The **profit margin** is the price minus marginal cost,  $P - MC$ . The **price markup** is the profit margin as a share of the price:

$$\text{markup} = \frac{P - MC}{P}$$

**Example:** the scooter company maximizes profit by producing where  $MR = MC$ , at 20 scooters, and setting  $P = 800$ . Its MC is 400.

- Its profit margin is  $800 - 400 = 400$  on each scooter.
- Its markup is  $400/800 = 1/2$ , or 50% of the price.

## The Markup Rule

When a firm maximizes profit, its markup equals one over the price elasticity of demand at its chosen point:

$$\frac{P - MC}{P} = \frac{1}{\varepsilon}$$

- **Example:** at  $E$ , the scooter demand curve has slope  $-20$ , so:

$$\varepsilon = -\frac{P}{Q} \times \frac{1}{\text{slope}} = \frac{800}{20} \times \frac{1}{20} = 2$$

- Then  $1/\varepsilon = 1/2$ , the same as its markup.
- The **less elastic** its demand, the **higher the markup** a firm sets.
- Fewer close substitutes make demand less elastic, so a firm with more market power sets a higher markup.

## Markups in the Data

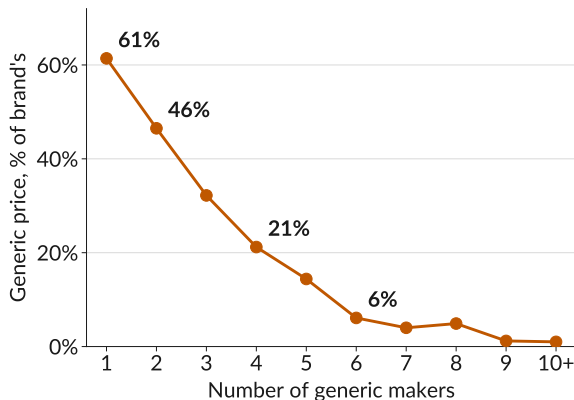
Estimates for US cars, with prices in 1983 dollars (Berry, Levinsohn, and Pakes, 1995):

| Car           | Price    | $P - MC$ | Markup | Elasticity |
|---------------|----------|----------|--------|------------|
| Mazda 323     | \$5,049  | \$801    | 16%    | 6.3        |
| Nissan Sentra | \$5,661  | \$880    | 16%    | 6.4        |
| Lexus LS400   | \$27,544 | \$9,030  | 33%    | 3.1        |
| BMW 735i      | \$37,490 | \$10,975 | 29%    | 3.4        |

- Small cars have many close substitutes, so their demand is elastic and their markups are low.
- Luxury cars have fewer substitutes, so their markups are about twice as high.

## Generic Drugs

- When a patent runs out, other firms can sell the same drug.
- FDA data: with one generic maker, the price is 61% of the brand's. With six, it is 6%.
- More sellers of the same drug means more elastic demand and lower prices.



## Where Does Market Power Come From?

A firm has market power when its buyers do not switch easily to rivals, so its demand is less elastic. We look at two sources of market power:

1. **Product differentiation**: the firm sells something different from its rivals, so buyers have fewer close substitutes.
2. **Cost advantages of size**: when producing more makes each unit cheaper, a few big firms can serve the whole market, and small rivals cannot keep up.

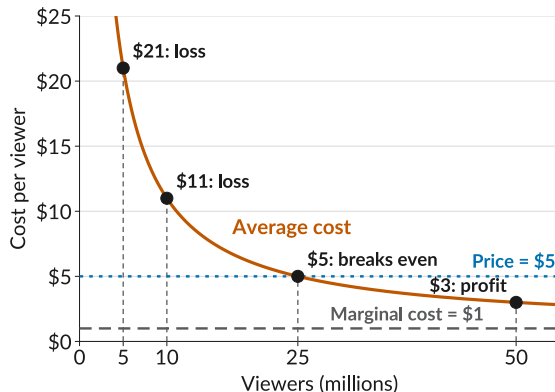
## Source 1: Product Differentiation

How do firms make their products different?

1. **Location and service:** shops differ by location, selection, and extras.
2. **Innovation:** the Toyota Prius, the first mass-produced hybrid (1997), had few close rivals for about 15 years.
3. **Advertising:** it informs buyers, draws them from rivals, and builds brand loyalty.
  - US cereal makers spend 5.5% of their revenue on ads.
  - Ads raised a cereal brand's demand more than price cuts did (Shum, 2004).

## Source 2: Cost Advantages of Size

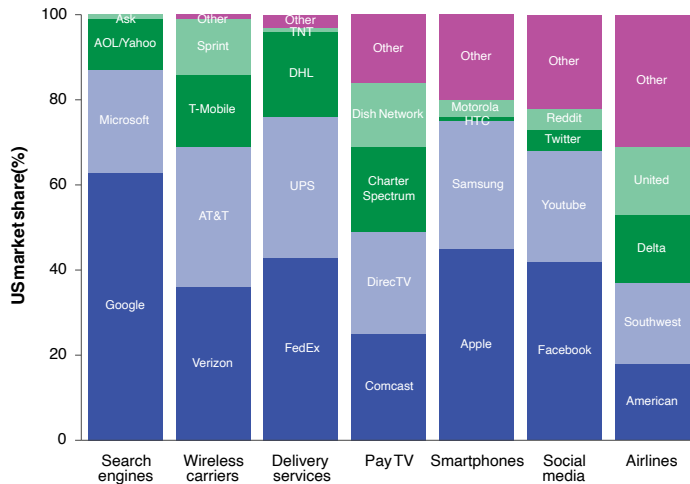
- A movie costs \$100 million to make and \$1 per extra stream, so AC falls as more people watch.
- At a \$5 price, a studio **breaks even** with 25 million viewers, and loses money with fewer.
- Two studios splitting 50 million viewers just break even. Three all lose. Only a few large firms survive.



## Natural Monopoly

- A **monopoly** is a firm that is the only seller of a product with no close substitutes.
- When a single firm can supply the whole market at a lower AC than two firms could, the market is a **natural monopoly**: water, electricity, and gas networks, whose pipes and wires are a huge fixed cost.
- With strongly falling AC, competition can be winner-takes-all. The more searches Google ran, the more data it gathered, and the cheaper it became to serve each user.

## Markets With Few Firms



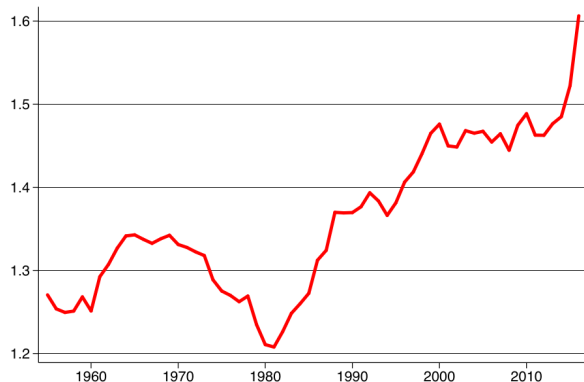
## Market Structures

Economists group markets by how many firms sell, how different their products are, and how easily new firms can enter:

| Structure                | Firms | Products       | Entry   | Control over price | Example           |
|--------------------------|-------|----------------|---------|--------------------|-------------------|
| Perfect competition      | Many  | Identical      | Easy    | None               | Wheat farms       |
| Monopolistic competition | Many  | Differentiated | Easy    | Some               | Restaurants       |
| Oligopoly                | Few   | Either         | Hard    | Some, rivals react | Wireless carriers |
| Monopoly                 | One   | No substitutes | Blocked | Most               | Water utility     |

## Markups Over Time

- The figure shows the average  $P/MC$  of US public firms (De Loecker, Eeckhout, and Unger, 2020).
- Prices went from 21% above MC in 1980 to 61% in 2016.
- In our measure,  $(P - MC)/P$ , that is 17% to 38%.



A price above MC means a deadweight loss. Should the government do something about rising market power?

## Competition Policy

**Competition policy** (or **antitrust policy**) is government policy and law that limits market power, prevents cartels, and regulates how firms compete.

- Authorities investigate when one or a few dominant firms are limiting their rivals, so buyers get less from competition than they could.
- They step in only when a firm has both the ability and the incentive to abuse its market power.
- Firms limit competition in two ways: by keeping rivals out, and by making their own demand less elastic.

## Keeping Rivals Out

- From expanding: UK supermarkets bought land near their stores and left it empty, so rivals could not build there.
- From inputs: Ferrero, the maker of Nutella, bought one of its hazelnut suppliers. Regulators allowed it, since enough other suppliers remained.
- From ideas: a **patent** gives a firm the sole right to an idea for a time, as a reward for research. When it expires, rivals enter: generic drugs.
- From buyers: Microsoft kept rival web browsers off Windows, so they could not reach its users. After years of investigation and fines, a browser market developed.

## Mergers, Cartels, and Loyalty

- Merging with rivals: Meta bought Instagram and WhatsApp. Authorities weigh the harm against the savings from scale and network effects, and have sometimes let such mergers through.
- Colluding: a **cartel** is a group of firms that agree to act like one monopoly. Cartels are illegal in most countries, but hard to prove.
- Making buyers less price-sensitive: advertising that builds a brand, paying for shelf space or the top spot in a search, and products that work best together, like Apple's.

## Case by Case

- Authorities weigh the harms and the benefits of each case on its own.
- For a natural monopoly, the answer is usually close regulation of its prices, or public ownership.
- Platforms are a dilemma: scale and network effects make them look like natural monopolies, yet many sellers and advertisers compete on them. The debate continues.
- Competition keeps moving, and policy often plays catch-up: Chrome now dominates the browser market that the Microsoft case opened up.

## What to Do Next

- Before next class: read sections 7.8, 7.9, 7.11, and 7.12, the markup rule in 7.6, and the **notes on market structures**. Go over the **slides**.
- We skip CORE's derivation of the markup rule from isoprofit curves, and its discussion of digital platforms.
- Work through the **practice problems** for this lecture.
- Next class: with only a few firms, each firm's best price depends on what its rivals charge. We model this as a game.